

Grades 9 & 10 Combined

Week of 2026-08-24

Grade 9: Expressions and Equations

1. Simplify: $3x + 5x$.
2. Simplify: $7a - 2a + 4$.
3. Evaluate $2x + 7$ when $x = 5$.
4. Solve: $x + 9 = 20$.
5. Solve: $4x = 28$.
6. Solve: $3x + 4 = 19$.
7. Solve: $5x - 7 = 18$.
8. Solve: $2(x + 3) = 16$.
9. Write an equation: Seven more than twice a number is 25.
10. A taxi charges \$4 plus \$2 per mile. Write a function $C(m)$ for the cost.

Grade 9: Linear Functions and Inequalities

11. Solve: $2x + 5 = 3x - 2$.
12. Solve: $4(x - 1) = 2x + 10$.
13. Solve and graph: $x + 3 > 7$.
14. Solve and graph: $2x - 5 \leq 9$.
15. Evaluate $f(x) = 3x - 2$ when $x = 4$.
16. Evaluate $g(x) = x^2 + 1$ when $x = 3$.
17. Is the relation $\{(1,2), (2,3), (3,4)\}$ a function?
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19. What is the slope of the line $y = 4x + 7$?
20. What is the y-intercept of the line $y = 3x - 5$?

Grade 9: Applications and Slope

21. A phone plan costs \$30 plus \$0.10 per text. Write an equation for the monthly cost C with t texts.
22. How much does the plan cost with 200 texts?
23. A student has \$45 and spends \$3 per notebook. Write an inequality for buying at least 10 notebooks.
24. Find the slope of the line through $(0, 3)$ and $(2, 11)$.
25. Explain what a slope of 5 means in the context of a car traveling 5 miles every hour.
26. Evaluate: 2^4 .
27. Evaluate: 3^3 .
28. Simplify: $x^3 \cdot x^2$.
29. Simplify: $x^6 \div x^2$.
30. Evaluate: $\sqrt{49}$.

Grade 10: Radicals and Equations

31. Write $\sqrt{x}$ using a rational exponent.
32. Write $x^{1/3}$ using a radical.
33. Evaluate: $16^{1/2}$.
34. Evaluate: $8^{1/3}$.
35. Simplify $\sqrt{36x^2}$ for positive x .

- 36. Simplify: $\sqrt{50}$.
- 37. Simplify: $\sqrt{72}$.
- 38. Simplify: $2\sqrt{3} + 5\sqrt{3}$.
- 39. Solve: $x^2 = 81$.
- 40. Solve: $(x - 4)^2 = 25$.

Grade 10: Transformations and Applications

- 41. Translate point (3, 2) right 4 units and up 1 unit. What are the new coordinates?
- 42. Reflect point (5, 3) across the x-axis. What are the new coordinates?
- 43. A triangle has vertices at (0, 0), (4, 0), and (0, 3). What is its area?
- 44. Describe the transformation from $y = x^2$ to $y = x^2 + 3$.
- 45. Describe the transformation from $y = x^2$ to $y = (x - 2)^2$.
- 46. Solve: $\sqrt{x + 5} = 4$.
- 47. The area of a square is 64 square units. What is the side length?
- 48. Simplify: $2\sqrt{12} + \sqrt{27}$.
- 49. Apply a dilation with scale factor 3 to point (2, -1). What are the new coordinates?
- 50. Explain why reflecting a point across the x-axis does not change the distance from the point to the x-axis.